

1 Solution — GIAM 5.3

Problem 2

1.1 Problem

Give an inductive proof that $\forall x \in \mathbb{N}, 3 \mid x^3 + 5x$.

1.2 Answer

$$(k+1)^3 + 5(k+1) = (k^3 + 5k) + 3(k^2 + k + 2)$$

Verified for every x up to 20000, with the increment checked as a polynomial identity.

1.3 The Proof

Theorem. For every natural number x , 3 divides $x^3 + 5x$.

Proof. By induction on x .

Basis. At $x = 0$ we have $0^3 + 5 \cdot 0 = 0$, and $3 \mid 0$.

Inductive step. Let k be a particular but arbitrarily chosen natural number and suppose $3 \mid k^3 + 5k$, say

$$k^3 + 5k = 3m \quad \text{for some integer } m.$$

Expanding the next case,

$$\begin{aligned}(k+1)^3 + 5(k+1) &= (k^3 + 3k^2 + 3k + 1) + (5k + 5) \\ &= (k^3 + 5k) + 3k^2 + 3k + 6.\end{aligned}$$

Substituting the hypothesis and factoring the rest,

$$= 3m + 3(k^2 + k + 2) = 3(m + k^2 + k + 2),$$

and $m + k^2 + k + 2$ is an integer. Hence $3 \mid (k + 1)^3 + 5(k + 1)$.

Basis and step together give the result for every natural number x . ■

1.4 The Picture

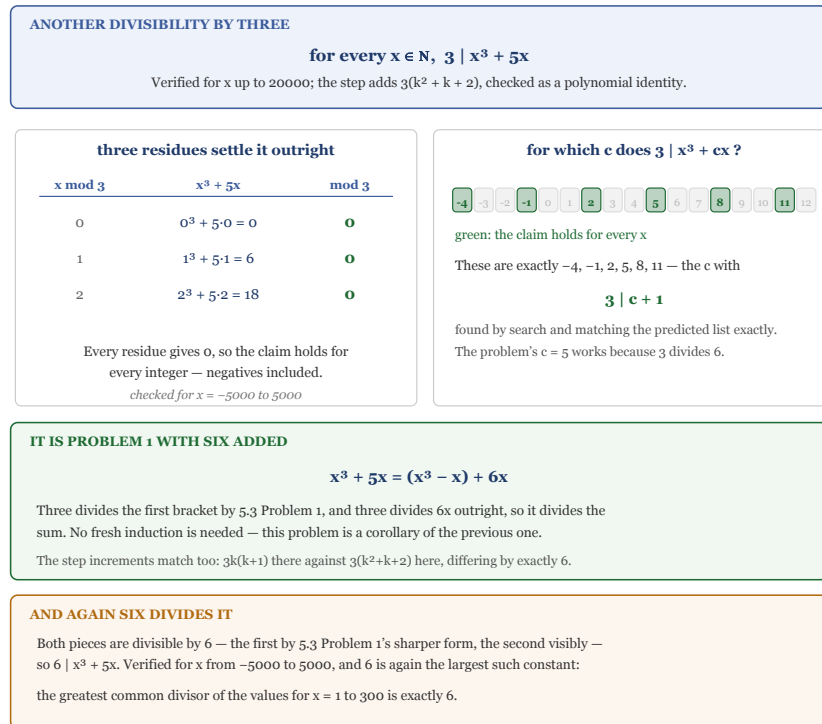


Figure 1.1: A diagram in four parts. The first states that for every natural number x , three divides x cubed plus five x , and notes it was verified for x up to 20000 with the step adding three times the quantity k squared plus k plus two, checked as a polynomial identity. The second part has two panels. The left panel shows three residues settling it outright: for x congruent to zero, x cubed plus five x is zero, which is zero mod three; for x congruent to one, it is one plus five equals six, zero mod three; for x congruent to two, it is eight plus ten equals eighteen, zero mod three. Every residue gives zero, so the claim holds for every integer including negatives, checked for x from minus 5000 to 5000. The right panel asks for which c the statement three divides x cubed plus $c x$ holds, showing a strip of c values from minus four to twelve with minus four, minus one, two, five, eight and eleven highlighted green.

These are exactly the c for which three divides c plus one, found by search

and matching the predicted list exactly, and the problem's c equals five works because three divides six. A green band explains that this is Problem 1 with six added: x cubed plus five x equals x cubed minus x , plus six x . Three divides the first bracket by 5.3 Problem 1 and divides six x outright, so it divides the sum, meaning no fresh induction is needed and this problem is a corollary of the previous one. The step increments match too, three k times k plus one there against three times k squared plus k plus two here, differing by exactly six. An orange band notes that again six divides it: both pieces are divisible by six, the first by Problem 1's sharper form and the second visibly, so six divides x cubed plus five x , verified for x from minus 5000 to 5000, and six is again the largest such constant since the greatest common divisor of the values for x from one to 300 is exactly six.

1.5 Remark — This Is Problem 1 with Six Added

The quickest route does not need a new induction at all:

$$x^3 + 5x = (x^3 - x) + 6x,$$

verified for x from -3000 to 3000 . The first bracket is divisible by 3 by 5.3 Problem 1, and $6x$ is divisible by 3 outright, so the sum is too.

The two inductive steps agree with this. Problem 1's increment was $3k(k+1) = 3k^2 + 3k$; this one is $3(k^2 + k + 2) = 3k^2 + 3k + 6$. They differ by exactly 6 , which is the derivative-like trace of having added $6x$ to the expression.

Worth noticing what the decomposition costs. Nothing is lost, and a reader who has seen Problem 1 can dispatch this one in a line. The induction written out above is the self-contained version, useful when the earlier result is not at hand but strictly longer.

1.6 Remark — Three Cases Prove It for Every Integer

An even shorter argument avoids both the induction and the previous problem. Every integer is $3q$, $3q+1$ or $3q+2$, and modulo 3 only the residue matters:

$x \bmod 3$	$x^3 + 5x$	$\bmod 3$
0	0	0
1	$1 + 5 = 6$	0
2	$8 + 10 = 18$	0

All three vanish, so $3 \mid x^3 + 5x$ — and this version covers **every integer**, negatives included, which the induction over $\mathbb{N}$ cannot reach. Confirmed for x from -5000 to 5000 .

This is the same trade met in 5.3 Problem 1. Induction walks upward from a basis and proves exactly the stated range; a residue check inspects finitely many cases and settles the whole of $\mathbb{Z}$ at once. For divisibility statements about a fixed modulus the residue check is usually the better tool, because the number of cases is the modulus rather than the size of x .

1.7 Remark — Which Coefficients Work

The decomposition generalises. For any integer c ,

$$x^3 + cx = (x^3 - x) + (c + 1)x,$$

and since $3 \mid x^3 - x$ always, the statement $3 \mid x^3 + cx$ holds for all x exactly when $3 \mid (c + 1)x$ for all x — that is, exactly when

$$3 \mid c + 1.$$

Searching c from -4 to 12 returns $-4, -1, 2, 5, 8, 11$, which is precisely the list of c with $c + 1$ divisible by 3 . The problem's $c = 5$ qualifies because $c + 1 = 6$.

So the 5 in the problem is not special; $2, 8$ or 11 would have served equally well, and $c = -1$ recovers Problem 1. By contrast $c = 0$ fails immediately, since $1^3 = 1$ is not a multiple of 3 .

1.8 Remark — Six Divides It Too

As in Problem 1 the stated divisor is not the best one. Both pieces of $(x^3 - x) + 6x$ are divisible by 6 — the first by Problem 1's sharper form, the second visibly — so

$$6 \mid x^3 + 5x \quad \text{for every integer } x,$$

verified from -5000 to 5000 . The values $0, 6, 18, 42, 84, 150, 246, \dots$ are all multiples of 6 , and the greatest common divisor of the values for $x = 1, \dots, 300$ is exactly 6 , so no larger constant works.

Two problems in a row where the structural argument gives a stronger conclusion than the induction is not an accident. The step propagates whatever property it was handed and no other, so an induction can only ever confirm the hypothesis it starts with — it has no mechanism for noticing that something better is true.